Mathematics

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Contents

Part I.

Algebra

1. Group Theory

1.1. Groups, permutations, subgroups

Definition 1.1.1. A *binary operation*, also referred to as a *law of composition* on a set S is a mapping from $S \times S$ to S. We say that a binary operation $\cdot\colon S \times S \to S$ is

- *associative* if $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in S$,
- *commutative* if $a \cdot b = b \cdot a$ for all $a, b, \in S$.

Remark 1.1.1. The $\cdot$ will usually be implicit. That is, ab will denote $a \cdot b$ where $\cdot$ is a binary operation.

The following are all familiar examples of laws of composition.

Example 1.1.1.

1. $+\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defines the mapping $(x, y) \mapsto x + y$.
2. $\cdot\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defines the mapping $(x, y) \mapsto xy$.
3. $+\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ defines the mapping $(a, b) \mapsto a + b$.
4. $M_{2\times 2}(\mathbb{R}) \times M_{2\times 2}(\mathbb{R}) \to M_{2\times 2}(\mathbb{R})$ has a mapping $(A, B) \mapsto AB$ (matrix multiplication).

All of the above examples were both associative and commutative, except for the last one, which was only associative (matrix multiplication does not commute in general).

Example 1.1.2. Let T be a set, and let M denote the set of functions from T to T. Take two elements f and g from M. Then there exists a law of composition $f \circ g$, known as function composition, defined as $(f \circ g)(t) \coloneqq f(g(t))$ for all $t \in T$. This law is associative, since $(f \circ (g \circ h))(t) = f(g(h(t))) = ((f \circ g) \circ h)(t)$.

Suppose we want to find the *product* of n elements $a_1 \cdots a_n$. If the law of composition is not associative, then there are many ways to write such a product, since the law of composition is only defined for two elements. For example, we could write the product as any of

$$\begin{aligned} &(a_1(a_2a_3))a_4\cdots \\ &(a_1a_2(a_3a_4))\cdots, \end{aligned}$$

which could all have distinct results. However, if the law is associative, then all ways will yield the same unique product.

Proposition 1.1.1. *Let there be given an associative law of composition on a set S. Then, there exists a unique way to write the product of n elements, which will temporarily be denoted as*

$$[a_1 \cdots a_n].$$

We define $[a_1 \cdots a_n]$ inductively with the following properties:

1. *The product of a single element is itself: $[a_1] = a_1$.*
2. *The product of two elements is given by the law of composition: $[a_1a_2] = a_1a_2$.*
3. *For all integers $1 \le i \le n-1$, we have $[a_1 \cdots a_n] = [a_1 \cdots a_i][a_{i+1} \cdots a_n]$.*

Proof. We will proceed via strong induction on n. The base case $n = 1$ and $n = 2$ are trivial. Now suppose that for all $k \le n-1$, we have already defined a product $[a_1 \cdots a_k]$. We must show that $[a_1 \cdots a_n]$ satisfies 3.

Take $[a_1 \cdots a_n] = [a_1 \cdots a_{n-1}][a_n]$. Both terms on the right hand side are unique by the hypothesis, so the left hand side must also be unique. Next, we have

$$\begin{aligned} [a_1 \cdots a_n] &= [a_1 \cdots a_{n-1}][a_n] \\ &= \big([a_1 \cdots a_j][a_{j+1} \cdots a_{n-1}]\big)[a_n], \end{aligned}$$

for all $1 \le j \le n-2$, and by associativity, we have

$$\begin{aligned} \big([a_1 \cdots a_j][a_{j+1} \cdots a_{n-1}]\big)[a_n] &= [a_1 \cdots a_j]\big([a_{j+1} \cdots a_{n-1}][a_n]\big) \\ &= [a_1 \cdots a_j][a_{j+1} \cdots a_n], \end{aligned}$$

by 3., and we are done. $\square$

Definition 1.1.2. An *identity* for a given law of composition $\cdot \colon S \times S \to S$ is an element $e \in S$ such that $ea = ae = a$ for all $a \in S$.

Remark 1.1.2. If the law of composition is written multiplicatively, then we will usually denote the identity element by 1. Otherwise, if it is written additively, we will usually denote it by 0.

Example 1.1.3. Multiplication has the identity 1, and addition has the identity 0. Matrix multiplication has the identity $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Function composition has the identity element $\mathrm{id} \colon X \to X$ such that $\mathrm{id}_X(x) = x$.

Proposition 1.1.2. *An identity element, if it exists, is unique.*

Proof. Let e and e' both be identities. Then, $ee' = e$ but also $ee' = e'e = e'$, so $e = e'$. □

Definition 1.1.3. Let there be a law of composition on S, and suppose that it has identity 1. An element $a \in S$ is *invertible* if there exists $b \in S$ such that $ab = ba = 1$. If a is invertible, we call b the *inverse* of a and write $a^{-1} = b$.

Proposition 1.1.3. *If a is invertible then a^{-1} is unique.*

Proof. Suppose b and b' are both inverses of a. Then $ab = ba = 1$ and $ab' = b'a = 1$. But then $b'ab = b' = (b'a)b = 1b = b$. Hence the inverse is unique. □

Remark 1.1.3. We denote the product of a with itself n times by $\underbrace{a \cdots a}_{n} = a^n$. By convention, a^0 is defined to be 1 (the identity). Finally, a^{-n} is the product of a^{-1} by itself n times. That is, $\underbrace{a^{-1} \cdots a^{-1}}_{n} = a^{-n}$.

Definition 1.1.4 (Group)**.** A *group* is a set G with a law of composition such that

1. the law of composition is associative,
2. G contains an identity element,
3. every element of G is invertible.

More formally, a group is an ordered pair of a set G and a binary operation $\cdot$ on G that satisfies the three group axioms.

Definition 1.1.5. If the law of composition on a group G is commutative, then G is said to be an *abelian* group.

Example 1.1.4.

- Addition is a group over $\mathbb{R}$. That is, $(\mathbb{R}, +)$ is a group. Moreover, it is abelian.
- $(\mathbb{Z}, \times)$ is not a group, because integers do not have multiplicative inverses.
- $(M_{2\times 2}(\mathbb{R}), \cdot)$ is not a group because some matrices are not invertible.
- Take $\mathbb{R}^\times := \mathbb{R} \setminus \{0\}$. Then $\mathbb{R}^\times$ is a group under multiplication.
- Let $\mathrm{GL}_n(\mathbb{R})$ be the set of $n \times n$ invertible matrices with real entries. Then $\mathrm{GL}_n(\mathbb{R})$ is a group under matrix multiplication. Note that it is not abelian, since matrix multiplication is not, in general, commutative.

Definition 1.1.6. The *order* of a group $(G, \cdot)$ is the number of elements in G. This is denoted by $|G|$. If $|G| < \infty$, then G is a *finite* group. Otherwise, it is an *infinite* group.

Proposition 1.1.4 (Cancellation law)**.** *Let $(G, \cdot)$ be a group, and take $a, b, c \in G$. Then, both of the following are true:*

1. *If $ab = ac$ or $ba = ca$, then $b = c$.*
2. *If $ab = a$ or $ba = a$, then $b = 1$.*

Proof.

1. Take $ab = ac$. Left multiplying by a^{-1} gives $a^{-1}b = a^{-1}c$ so $b = c$. The other case follows identically.
2. Take $c = 1$ in part 1.

□

Definition 1.1.7. A *permutation* is a bijection $\sigma\colon S \to S$. The set of permutations of a set X is denoted S_X and called the *symmetric group* of X. In particular, if $X = \{1, 2, \ldots, n\}$ and $|X| = n$, we write S_n and say that it is the symmetric group of degree n.

Proposition 1.1.5. *The symmetric group S_n is a group under function composition. That is, $(S_n, \circ)$ is a group.*

Proof. We must check that function composition is closed and associative, all elements are invertible and that there exists an identity.

1. Obviously, the composition of two bijections is also a bijection. Hence S_n is closed under $\circ$.

2. It is also obvious that function composition is associative (we proved this earlier).

3. Since all bijections are invertible, we know that all elements of S_n have an inverse.

4. Finally, the identity function id which maps every element to itself is the identity.

Hence $(S_n, \circ)$ is a group. $\square$

Remark 1.1.4. The order of S_n is $|S_n| = n!$.

Example 1.1.5. Take $\sigma \in S_5$ defined according to the table

i	1	2	3	4	5
$\sigma(i)$	2	3	1	5	4

We will write this as $\sigma = (123)(45)$, with each bracketed string forming a 'cycle'. In particular, 2-cycles like (45) are called *transpositions*. Note that cycle notation is not unique. In cycle notation, if an element is omitted, it is assumed that it maps to itself.

Example 1.1.6. Take $s, t \in S_5$, and let $s = (123)(45)$ and $t = (12)(34)$. Then $st = s \circ t = (3541)$. Similarly, $ts = t \circ s = (2453)$. Note that $st \neq ts$ so S_n is not abelian.

Every permutation has an associated permutation matrix. The idea is that for every $s \in S_n$, there exists $P_s \in M_{n \times n}$ such that $P_s A$ permutes the rows of A according to s. This is trivially constructed by applying the permutation to *columns* of the identity matrix. We have

$$(P_s)_{ij} = \begin{cases} 1 & s(j) = i \\ 0 & \text{otherwise.} \end{cases}$$

For example, consider the permutation $s = (123)(45)$. Take the standard basis vectors $\mathbf{e_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and so on. We want $P_s\mathbf{e_1} = \mathbf{e_2}$, $P_s\mathbf{e_2} = \mathbf{e_3}$, $P_s\mathbf{e_3} = \mathbf{e_1}$ and so on. By permuting the columns, we get

$$P = \left[\mathbf{e_2} \,|\, \mathbf{e_3} \,|\, \mathbf{e_1} \,|\, \mathbf{e_5} \,|\, \mathbf{e_4}\right] = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

In fact, matrix multiplication by permutation matrices results in the composition of the permutations.

Definition 1.1.8. Let P_s be the permutation matrix associated with $s \in S_n$. The *sign* of s is defined to be $\operatorname{sgn}(s) = \det(P_s) = \pm 1$. A permutation is *even* if its sign is 1, and *odd* if its sign is -1.

Remark 1.1.5. Every permutation is a product of transpositions. This is equivalent to the fact that the permutation matrix is the result of a series of elementary row operations on the identity matrix.

Definition 1.1.9 (Subgroup)**.** A subset H of a group G is a *subgroup* if H is a group with respect to the law of composition of G.

Example 1.1.7.

1. $\mathbb{R}^\times$ is a subgroup of $\mathbb{C}^\times$ under multiplication.
2. $\{z \in \mathbb{C} : |z| = 1\}$ is a subgroup of $\mathbb{C}^\times$. This is the *circle group*.
3. $\{A \in \mathrm{GL}_n(\mathbb{R}) : \det(A) = 1\}$ is a subgroup of $\mathrm{GL}_n(\mathbb{R})$. In fact, this subgroup is so important that we give it a name, the *special linear group*, and denote it with $\mathrm{SL}_n(\mathbb{R})$.

Remark 1.1.6. If G is a group, then it contains two obvious subgroups: G itself and $\{1\}$ (the *trivial group*).

Definition 1.1.10. A subgroup H of G is a *proper* subgroup if $H \neq G$.

With this definition, the trivial subgroup is a proper subgroup of any non-trivial group.

Take a positive integer a. Then, the set of all multiples of a is denoted by

$$a\mathbb{Z} \coloneqq \{x \in \mathbb{Z} : x = ak, \text{ where } k \in \mathbb{Z}\}.$$

Proposition 1.1.6. *The set $a\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ under addition.*

Proposition 1.1.7. *TODO*

Theorem 1.1.1. *Let S be a subgroup of $(\mathbb{Z}, +)$. Then either $S = \{0\}$ or $S = a\mathbb{Z}$ for some positive integer a.*

Proof. Suppose that S is non-trivial, and contains some element other than 0. □

Take two positive integers a and b and define the set

$$a\mathbb{Z} + b\mathbb{Z} \coloneqq \{ar + bs : r, s \in \mathbb{Z}\}.$$

Proposition 1.1.8. *The set $a\mathbb{Z} + b\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ under addition.*

Proof. TODO □

Since $a\mathbb{Z} + b\mathbb{Z}$ is a (non-trivial) subgroup of $\mathbb{Z}$, by Theorem 1.1.1, it must be equal to some $d\mathbb{Z}$.

Definition 1.1.11. Let d be the unique positive integer such that $a\mathbb{Z}+b\mathbb{Z} = d\mathbb{Z}$. We call d the *greatest common divisor* of a and b, denoted $d = \gcd(a, b)$.

Proposition 1.1.9. *The greatest common divisor defined in this way has the same properties as the usual definition of the greatest common divisor. That is, if $d = \gcd(a, b)$, then*

1. *$d \mid a$ and $d \mid b$,*
2. *if $e \in \mathbb{Z}$ and $e \mid a$ and $e \mid b$, then $e \mid d$,*
3. *there exists $r, s \in Z$ such that $ar + bs = d$.*

Proof. TODO □

Proposition 1.1.10 (Euclidean algorithm)**.** *Let n, d be natural numbers such that $n = qd + r$. Then, $\gcd(n, d) = \gcd(d, r)$.*

Proof. We will prove that the set of common divisors of n and d is the same as the set of common divisors of d and r.

Suppose a is a common divisor of n and d, that is, $a \mid n$ and $a \mid d$. This means that $a \mid n\alpha + d\beta$ for any integers α, β. Specifically, we have $a \mid n - qd = r$, so $a \mid r$. Therefore a is a common divisor of n and r.

Similarly, suppose a is a common divisor of d and r, that is, $a \mid d$ and $a \mid r$. We have $a \mid qd + r = n$, so $a \mid n$. Therefore a is a common divisor of n and d.

The set of divisors are therefore the same, so the greatest common divisor must also be the same. □

Example 1.1.8. We can find the greatest common divisor of two numbers using the Euclidean algorithm. For example, let $a = 154$ and $b = 62$. Since $a = 2b + 30$, we have

$$\begin{aligned} a\mathbb{Z} + b\mathbb{Z} &= (2b + 30)\mathbb{Z} + b\mathbb{Z} \\ &= b\mathbb{Z} + 30\mathbb{Z}. \end{aligned}$$

Now $b = 2 \times 30 + 2$, so

$$\begin{aligned} b\mathbb{Z} + 30\mathbb{Z} &= (2 \times 30 + 2)\mathbb{Z} + 30\mathbb{Z} \\ &= 30\mathbb{Z} + 2\mathbb{Z}. \end{aligned}$$

Now the greatest common divisor is obviously 2. Hence $d = 2$.

Definition 1.1.12. Two non-zero integers a, b are *coprime*, or *relatively prime*, if $\gcd(a, b) = 1$. That is, $a\mathbb{Z} + b\mathbb{Z} = \mathbb{Z}$.

Proposition 1.1.11. *Two non-zero integers a and b are coprime if and only if there exist r, s such that $ar + bs = 1$.*

Proof. TODO $\square$

Proposition 1.1.12 (Euclid's lemma)**.** *Let p be a prime number, and let a, b be non-zero integers. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. TODO $\square$

Proposition 1.1.13. *Let a, b be non-zero integers. Then $a\mathbb{Z} \cap b\mathbb{Z}$ is a subgroup.*

Proof. TODO $\square$

Hence by Theorem 1.1.1, $a\mathbb{Z} \cap b\mathbb{Z} = m\mathbb{Z}$ for some integer m.

Definition 1.1.13. For any two non-zero integers a and b, the *least common multiple* of a and b is m such that $a\mathbb{Z} \cap b\mathbb{Z} = m\mathbb{Z}$. We denote m by $\operatorname{lcm}(a, b)$.

Proposition 1.1.14. *Let a, b be non-zero integers and let $m = \operatorname{lcm}(a, b)$. Then,*

1. *$a \mid m$ and $b \mid m$,*
2. *if $a \mid n$ and $b \mid n$, then $m \mid n$.*

Proof. TODO $\square$

Corollary 1.1.1. *Let $d = \gcd(a, b)$ and $m = \gcd(a, b)$ for two non-zero integers a, b. Then $ab = dm$.*

1.2. Cyclic groups

Definition 1.2.1. Let G be a group and take $x \in G$. The *cyclic subgroup* of G genearted by x is

$$\langle x \rangle := \{x^k : k \in \mathbb{Z}\}$$

Proposition 1.2.1. *Let G be a group and take $x \in G$. Let $\langle x \rangle$ be the cyclic subgroup generated by x. Suppose $S = \{k \in \mathbb{Z} : x^k = 1\}$. Then,*

1. *the set S is a subgroup of $(\mathbb{Z}, +)$,*
2. *$x^r = x^s$ if and only if $x^{r-s} = 1$*

Proof. TODO $\square$

2. Linear Algebra

2.1. Vector spaces

We begin by recalling the definition of vector spaces.

Definition 2.1.1 (Vector space)**.** Let $\mathbb{F}$ be a field. A *vector space* over $\mathbb{F}$ is a set V with a binary operation $+$ and function $\cdot\colon \mathbb{F} \times V \to V$ such that

- $(V, +)$ is an abelian group with identity $\mathbf{0}$,
- $1v = v$ for all $v \in V$ (identity),
- $a(bv) = (ab)v$ for all $a, b \in \mathbb{F}$ and $v \in V$ (associativity),
- $a(u + v) = au + bv$ and $(a + b)v = av + bv$ for all $a, b \in \mathbb{F}$ and $u, v \in V$ (distributivity).

Example 2.1.1. Common examples of vectors spaces include

- $\mathbb{F}^n = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{F}\}$ (e.g. $\mathbb{R}^2$)
- $M_{m\times n}(\mathbb{F})$, the set of $m \times n$ matrices with entries in $\mathbb{F}$
- $P_n(\mathbb{F})$, the set of polynomials of degree n with coefficients in $\mathbb{F}$
- $C(\mathbb{R}, \mathbb{R})$, the set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$
- $C^1(\mathbb{R}, \mathbb{R})$, the set of differentiable functions from $\mathbb{R}$ to $\mathbb{R}$

Linear algebra is the study of linear transformations — maps that preserve vector addition and scalar multiplication.

Definition 2.1.2 (Linear transformation)**.** Let V, W be vector spaces over $\mathbb{F}$. A *linear transformation*, or *linear map*, is a function $T\colon V \to W$ such that:

- $T(u + v) = T(u) + T(v)$ for all $u, v \in V$,
- $T(\alpha v) = \alpha T(v)$ for all $\alpha \in \mathbb{F}$, $v \in V$.

A linear transformation from V to itself is a *linear operator*.

Indeed, linear transformations are a kind of homomorphism. To this end, we will let $\operatorname{Hom}(V, W)$ denote the set of linear transformations from V to W. (This is also sometimes notated as $\mathcal{L}(V, W)$.)

Proposition 2.1.1. *The set of linear transformations from V to W, $\operatorname{Hom}(V, W)$ is a vector space.*

Proof. We first show that linear transformations form an abelian group under multiplication. Obviously there is an additive identity, the zero map that takes every element to the zero vector. Let $S, T \in \operatorname{Hom}(V, W)$. Then define $(S+T)(v) := S(v) + T(v)$. This is obviously a linear map. It is also not hard to verify that scalar multiplication works as it should in a vector space, and that distributivity holds. □

Definition 2.1.3. A linear transformation $T \colon V \to W$ is *invertible* if there exists a linear transformation $S \colon W \to V$ such that $TS = I_V$ and $ST = I_W$.

Proposition 2.1.2. *A linear transformation is invertible if and only if it is bijective.*

Proof. TODO □

Definition 2.1.4. A linear transformation that is bijective is an *isomorphism*.

Proposition 2.1.3. *Let U, U', V, V' be vector spaces over $\mathbb{F}$, and suppose that $\varphi \colon U \to U'$ and $\psi \colon V \to V'$ are isomorphisms. Then the map $\Phi \colon \operatorname{Hom}(U, V) \to \operatorname{Hom}(U', V')$ where $\Phi = \psi \circ T \circ \varphi^{-1}$ is an isomorphism.*

Proof. We need to show that Ψ is both a linear transformation and bijective. Also refer to the commutative diagram [TODO] □

We know that every linear transformation has a corresponding matrix. This is because every linear transformation is completely defined by its values on a generating (spanning) set. That is, suppose $S \subseteq V$ is a set that spans V. This means that every vector in V is a linear combination of the vectors in S. Then the values that a linear transformation $T \colon V \to W$ takes for all $s \in S$ completely define it. In other words, if we know $T(s)$ for all $s \in S$ where S spans V, then we know T.

Definition 2.1.5. A set $S \subseteq V$ is said to *span* V if any v in V can be written as a linear combination of vectors in S. That is, if $S = \{s_1, s_2, \ldots, s_n\}$, then for any $v \in V$ we can write

$$v = \alpha_1 s_1 + \cdots + \alpha_n s_n$$

for scalars $\alpha_i \in \mathbb{F}$.

Definition 2.1.6. Let $S = \{s_1, s_2, \dots, s_n\} \subseteq V$. Then S is said to be *linearly independent* if

$$\alpha_1 s_1 + \cdots + \alpha_n s_n = \mathbf{0}$$

if and only if $\alpha_i = 0$ for all $1 \le i \le n$.

Definition 2.1.7. A *basis* for a vector space V is a linearly independent set that spans V.

Remark 2.1.1. We only deal with *finite* dimensional vector spaces here, so all our bases have a finite number of elements. In addition, our definition is technically a *Hamel* basis, as there are a number of ways to define a basis.

Proposition 2.1.4. *Any two bases of a vector space have the same number of elements.*

Proof. TODO ☐

Definition 2.1.8. The *dimension* of a vector space V is the number of elements in a basis of V, and is denoted $\dim(V)$.

Bases are a useful way to assign coordinates to vectors. In particular, if $\dim(V) = n$, then $\mathbb{F}^n$ is isomorphic to V, where V is a vector space. The isomorphism is formed by choosing a basis of V.

Suppose that V is a vector space and let B be a basis of V. If $v \in V$, then we write the *coordinate vector* of v with respect to B as $[v]_B$ or $B^{-1}v$.

If $B = \{v_1, v_2, \dots, v_n\}$ and $v = \sum_{i=1}^n \alpha_i v_i$, then we can write the column vector

$$[v]_B = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}.$$

In fact, we can define (by abuse of notation) the map $B\colon \mathbb{F}^n \to V$ where $(x_1, \dots, x_n) \mapsto x_1 v_1 + \cdots + x_n v_n$.

Now consider a linear map $T\colon U \to V$, where $\dim U = n$ and $\dim V = m$. Let B and B' be bases for U and V respectively. Then $(B')^{-1} \circ T \circ B \in \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m) \cong M_{m\times n}(\mathbb{F})$.

Let's see what's going on here. We basically are mapping an element of $\mathbb{F}^n$ to an element of $\mathbb{F}^m$ by the following process:

$$\mathbb{F}^n \xrightarrow{B} U \xrightarrow{T} V \xrightarrow{(B')^{-1}} F^m.$$

Moreover, this map is a linear map since it is the composition of a bunch of linear maps (B, B' are isomorphisms), so $(B')^{-1} \circ T \circ B$ is also a linear map from $\mathbb{F}^n$ to $\mathbb{F}^m$. We also know that maps from $\mathbb{F}^n$ to $\mathbb{F}^m$ are represented as matrices. In particular, if $U = V$, then we will write $B^{-1} \circ T \circ B = [T]_B$.

Example 2.1.2. Let $T\colon P_2(\mathbb{F}) \to P_2(\mathbb{F})$ be the map defined by $T(p(x)) = \frac{d}{dx}p(x)$. Then,

$$\frac{d}{dx}(1) = 0$$
$$\frac{d}{dx}(x) = 1$$
$$\frac{d}{dx}(x^2) = 2x$$

so

$$[T]_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Part II.

Analysis

3. Real Analysis

Part III.

Probability

4. Probability

4.1. Basics

Definition 4.1.1. The *sample space*, denoted Ω is the set of all possible outcomes of an experiment.

Each element $\omega \in \Omega$ is called an *outcome*. An *event* E is a subset of Ω, and is hence a set of possible outcomes. Thus Ω itself is an event, the *certain event*, and so is the empty set $\varnothing$, the *impossible event*. An event E *occurs* if the outcome ω is in E.

Definition 4.1.2. If Ω is countable, then we say that it is a *discrete* sample space. Otherwise, if Ω is uncountable, we say that it is a *continuous* sample space.

Since events are sets, we can use the usual properties of sets. In particular, if A and B are events, then $A \cup B$ is an event (that either A or B occur) and $A \cap B$ is an event (that both A and B occur).

Proposition 4.1.1 (Set theory properties)**.** *Let A, B, C be subsets of Ω.*

1. *We have $A \cup \varnothing = A$ and $A \cap \varnothing = \varnothing$.*

2. *We have $A \cup \Omega = \Omega$ and $A \cap \Omega = A$.*

3. *We have $A \cup (\Omega \setminus A) = \Omega$ and $A \cap (\Omega \setminus A) = \varnothing$. In other words, $A \cup A^c = \Omega$ and $A \cap A^c = \varnothing$.*

4. *(Commutativity) We have $A \cup B = B \cup A$ and $A \cap B = B \cap A$.*

5. *(Associativity) We have $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$.*

6. *(Distributivity) We have $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.*

7. *(De Morgan's laws) We have $\Omega \setminus (A \cup B) = (\Omega \setminus A) \cap (\Omega \setminus B)$ and $\Omega \setminus (A \cap B) = (\Omega \setminus A) \cup (\Omega \setminus B)$. In other words, $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.*

Definition 4.1.3. Two events A and B are *mutually exclusive* or *disjoint* if $A \cap B = \varnothing$. A collection of events $\{E_i\}_{i \in I}$ is mutually exclusive if $E_i \cap E_j = \varnothing$ for all $i \neq j$.

Definition 4.1.4. Let Ω be a sample space and let 2^Ω denote the power set of Ω. Then $\mathbb{P} \colon 2^\Omega \to [0,1]$ is a probability function that assigns to each event a probability. This function $\mathbb{P}$ must satisfy the following axioms formulated by Kolmogorov:

1. For all events E, we have $\mathbb{P}(E) \geq 0$.
2. The probability of the sample space is 1. That is, $\mathbb{P}(\Omega) = 1$.
3. (Countable additivity) Let $E_1, E_2, \ldots$ be a countable sequence of mutually exclusive (disjoint) events. Then,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i).$$

Remark 4.1.1. This differs from the definition of probability space one will encounter later on in a more advanced, measure theoretic course. The formal definition of a probability space is the following: a probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F} \subseteq 2^\Omega$ is a σ-algebra, and $\mathbb{P}$ is a probability measure $\mathbb{P} \colon \mathcal{F} \to [0,1]$ satisfying $\mathbb{P}(A) \geq 0$, $\mathbb{P}(\Omega) = 1$, and countable additivity.

In particular, our definition does not work for some uncountable sample spaces. The restriction for $\mathcal{F}$ to be a σ-algebra is exactly what is needed to 'fix' this issue.

Proposition 4.1.2 (Properties of probability)**.** *Let Ω be a sample space and let A, B be events. Let $\mathbb{P}$ be a probability function on Ω.*

1. *We have $\mathbb{P}(\varnothing) = 0$.*
2. *We have $\mathbb{P}(A^c) = \mathbb{P}(\Omega \setminus A) = 1 - \mathbb{P}(A)$.*
3. *We have $\mathbb{P}(A) \leq 1$.*
4. *We have $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.*
5. *If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*
6. *(Finite additivity) If $E_1, E_2, \ldots, E_n$ are mutually exclusive events, then*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} \mathbb{P}(E_i)$$

7. *If* $A_1 \subseteq A_2 \subseteq \dots$ *and* $B = \bigcup_{i=1}^{\infty} A_i$*, then* $\mathbb{P}(B) = \lim_{n\to\infty} \mathbb{P}(A_n)$*.*

8. *If* $A_1 \supseteq A_2 \supseteq \dots$ *and* $B = \bigcap_{i=1}^{\infty} A_i$*, then* $\mathbb{P}(B) = \lim_{n\to\infty} \mathbb{P}(A_n)$*.*

Proof. We will prove (1), (2), (4), (6), and (7), and leave the rest as exercises.

1. This is a corollary of (2). Since $\varnothing = \Omega^c$, we have $\mathbb{P}(\varnothing) = \mathbb{P}(\Omega^c) = 1 - \mathbb{P}(\Omega) = 1 - 1 = 0$.

2. This is a corollary of finite additivity (6), with $n = 2$. Consider the disjoint events A and A^c. Since $A \cup A^c = \Omega$, we have $1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c)$, from which the result follows.

3. Exercise.

4. Again, this follows from finite additivity applied to the disjoint sets A and $B \setminus A$. Since $A \cup (B \setminus A) = A \cup B$, we have

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A \cup (B \setminus A)) \\ &= \mathbb{P}(A) + \mathbb{P}(B \setminus A). \end{aligned}$$

   Notice that $B = (B \setminus A) \cup (A \cup B)$ is a disjoint union, from which we get $\mathbb{P}(B) = \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B)$, so $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. This, combined with the previous equation, yields

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

5. Exercise.

6. Consider the mutually exclusive events $E_1, \dots, E_n$. Let $E_{n+1} = E_{n+2} = \dots = \varnothing$. Notice that $E_1, \dots$ still form a mutually exclusive collection of events, so countable additivity applies. Thus

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(E_i) \\ &= \sum_{i=1}^{n} \mathbb{P}(E_i) + \sum_{i=n+1}^{\infty} \mathbb{P}(E_i) \\ &= \sum_{i=1}^{n} \mathbb{P}(E_i) + 0, \end{aligned}$$

   since $\mathbb{P}(E_i) = \mathbb{P}(\varnothing) = 0$ for all $i > n$. Hence finite additivity holds.

7. Define $B_i = A_i \setminus A_{i-1}$. Clearly all of B_i are mutually exclusive, and moreover, $B = \bigcup_{i=1}^{\infty} B_i$. Hence by countable additivity,

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \sum_{i=1}^{\infty} \mathbb{P}(B_i) \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(B_i) \\ &= \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{n} B_i\right),\end{aligned}$$

with the last step being due to finite additivity. But $\bigcup_{i=1}^{n} B_i = A_n$, so we have

$$\mathbb{P}(B) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

as required.

8. Exercise. □

In a discrete sample space, all outcomes are disjoint. (This is obviously true, due to the fact that there can be only one outcome of an experiment.) Hence for any event E, we have that

$$\mathbb{P}(E) = \sum_{\omega\in E} \mathbb{P}(\{\omega\}).$$

Remark 4.1.2. It is important to note that the statements $\mathbb{P}(E) = 0 \implies E = \varnothing$ and $\mathbb{P}(E) = 1 \implies E = \Omega$ are **not necessarily true**. For example, let $\Omega = \{H, T\}$ and define $\mathbb{P}(H) = 0$ and $\mathbb{P}(T) = 1$. Notice that $H \neq \varnothing$ and $T \neq \Omega$.

4.2. Conditional probability and independence

Definition 4.2.1. The *conditional probability* of event A *given* event B is defined as

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

assuming that $\mathbb{P}(B) > 0$.

This definition may be rearranged to get the equivalent formulation

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B),$$

which sometimes may be more useful.

Definition 4.2.2. We say that events A and B are *positively related* (or *positively correlated*) if $\mathbb{P}(A \mid B) > \mathbb{P}(A)$. Similarly, events A and B are *negatively related* if $\mathbb{P}(A \mid B) < \mathbb{P}(A)$.

A simple calculation will show that if $\mathbb{P}(A \mid B) > \mathbb{P}(A)$, then $\mathbb{P}(B \mid A) > \mathbb{P}(B)$. Hence the intuition is that if two events are positively related, then knowing one event will boost the likelihood of the other event.

Definition 4.2.3. Two events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

If A and B are independent, then $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, and similarly, $\mathbb{P}(B \mid A) = \mathbb{P}(B)$. However, these equations involving conditional probability only make sense when $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Hence we have defined independence to make sense even when some of the probabilities are zero.

Proposition 4.2.1 (Properties of independence)**.** *Suppose that A and B are independent events.*

1. *We have that A and B^c are independent.*
2. *We have that A^c and B are independent.*
3. *We have that A^c and B^c are independent.*

Proof.

1. Since A and B are independent, we have $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$. Since $A = (A \cap B) \cup (A \cap B^c)$ is a disjoint union, we have

$$\begin{aligned}
\mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \\
\mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\
&= \mathbb{P}(A) - \mathbb{P}(A)\,\mathbb{P}(B) \\
&= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\
&= \mathbb{P}(A)\,\mathbb{P}(B^c),
\end{aligned}$$

so we see that A and B^c are independent.

2. Follows directly from (1) by swapping A and B.

3. By de Morgan, we have $A^c \cap B^c = (A \cup B)^c$. Hence

$$\begin{aligned}\mathbb{P}(A^c \cap B^c) &= 1 - \mathbb{P}(A \cup B) \\ &= 1 - (\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)) \\ &= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A)\,\mathbb{P}(B) \\ &= (1 - \mathbb{P}(A))(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A^c)\,\mathbb{P}(B^c),\end{aligned}$$

   so A^c and B^c are independent. □

We can extend our definition of independence to multiple events.

Definition 4.2.4. Events $E_1, E_2, \ldots, E_n$ are *independent* or *mutually independent* if for every $k \leq n$ and $1 \leq i_1 \leq \cdots \leq i_k \leq n$,

$$\mathbb{P}\left(\bigcap_{j=1}^{k} E_{i_j}\right) = \prod_{j=1}^{k} \mathbb{P}(E_{i_j}).$$

Remark 4.2.1. This definition is different from that of *pairwise* independence, which only requires that every pair of events is independent. Mutual independence is strictly stronger, in the sense that mutual independence implies pairwise independence, but not the converse.

Definition 4.2.5. We say that a collection of events $\{E_i\}_{i \in I}$ is *exhaustive* if

$$\bigcup_{i \in I} E_i = \Omega.$$

In particular, note that A and A^c are exhaustive, and also disjoint.

Definition 4.2.6. A *partition* of the sample space Ω is a collection of events that are mutually exclusive and exhaustive. That is, if $\{E_i\}_{i \in I}$ is a collection of events, then it is a partition if and only if:

1. (Mutually exclusive) For all $i \neq j$, $E_i \cap E_j = \varnothing$.
2. (Exhaustive) We have $\bigcup_{i \in I} E_i = \Omega$.

A partition $\{E_i\}_{i \in I}$ is countable if the set is countable, and uncountable if the set is uncountable.

Proposition 4.2.2 (Law of total probability)**.** *If $\{E_i\}_{i\in I}$ is a countable partition of Ω, then for any event A,*

$$\mathbb{P}(A) = \sum_{i\in I} \mathbb{P}(A \cap E_i).$$

or equivalently,

$$\mathbb{P}(A) = \sum_{i\in I} \mathbb{P}(A \mid E_i)\,\mathbb{P}(E_i).$$

Proposition 4.2.3 (Bayes' formula)**.** *Given events A and B, with $\mathbb{P}(B) > 0$, we have*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Equivalently, if $\{E_i\}_{i\in I}$ is a countable partition of Ω, and A is an event with $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(E_k \mid A) = \frac{\mathbb{P}(A \mid E_k)\,\mathbb{P}(E_k)}{\sum_{i\in I} \mathbb{P}(A \mid E_i)\,\mathbb{P}(E_i)}.$$

4.3. Random variables

Definition 4.3.1. A *random variable* X is a function $X\colon \Omega \to \mathbb{R}$ that assigns a real number to every outcome in the sample space.

We define the probability of X taking on a value x as

$$\mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}).$$

In general, for any subset $A \subseteq \mathbb{R}$, we have

$$\mathbb{P}(X \in A) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\}).$$

Definition 4.3.2. The *state space* of X, the set of possible values that X can take, is denoted S_X. If S_X is countable, then X is a *discrete* random variable.

Remark 4.3.1. One can have a discrete random variable in an uncountable sample space. For example, let $\Omega = \mathbb{R}$ and define

$$X = \begin{cases} 1 & \text{if } \omega \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 4.3.2. A random variable partitions Ω. This is the point of having random variables — to carve up the sample space into smaller chunks, to simplify calculations.

Definition 4.3.3. The *probability mass function* (or *pmf*) of a discrete random variable X is a function $p_X : S_X \to [0,1]$ defined by

$$p_X(x) = \mathbb{P}(X = x).$$

The *cumulative distribution function* or *distribution function* or *cdf* of any random variable X is a function $F_X : \mathbb{R} \to [0,1]$ defined by

$$F_X(x) = \mathbb{P}(X \leq x).$$

Proposition 4.3.1. *The pmf p_X of a discrete random variable X satisfies:*

1. *For all $x \in S_x$, $p_X(x) \geq 0$.*
2. *The sum over all x in the state space is $\sum_{x \in S_X} p_X(x) = 1$.*

Proposition 4.3.2 (Properties of cdfs)**.** *The cdf F_X of a random variable X satisfies:*

1. *For real numbers $a < b$, we have $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$.*
2. *F_X is non-decreasing (increasing).*
3. *$\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$.*
4. *F_X is right-continuous. That is, $\lim_{x \to y^+} F_X(x) = F_X(y)$ for all $y \in \mathbb{R}$.*

Proposition 4.3.3. *A function G is the cdf of a random variable if and only if G is non-decreasing, right-continuous, and has $\lim_{x \to -\infty} G(x) = 0$ and $\lim_{x \to \infty} G(x) = 1$.*

Definition 4.3.4. A random variable X is *continuous* if its distribution function F_X is continuous (in particular, it is left-continuous as well as right-continuous). In this case, S_X will be uncountable.

Proposition 4.3.4. *Let X be a random variable. Then,*

$$\mathbb{P}(X < x) = \lim_{t \to x^-} F_X(t)$$

and

$$\mathbb{P}(X = x) = F_X(x) - \lim_{t \to x^-} F_X(t).$$

In particular, $P(X = x) = 0$ for all $x \in S_X$ if and only if F_X is continuous.

Proof. The second equation $\mathbb{P}(X = x) = F_X(x) - \lim_{t\to x^-} F_X(t)$ immediately follows from the first, due to the fact that $\mathbb{P}(X = x) = \mathbb{P}(X \le x) - \mathbb{P}(X < x)$.

Consider $\mathbb{P}(X \le x-h) = F_X(x-h)$ for some real number $h > 0$. Then $\mathbb{P}(X < x) = \lim_{h\to 0} \mathbb{P}(X \le x - h) = \lim_{h\to 0} F_X(x - h) = \lim_{t\to x^-} F_X(t)$, as required. (Formally, we would use $A_i \coloneqq \{X \le x - \frac{1}{n}\}$ and apply Proposition 4.1.2(8).)

Now we see that $\mathbb{P}(X = x) = 0$ if and only if $F_X(x) = \lim_{t\to x^-} F_X(t)$, which is equivalent to left-continuity. But since all distribution functions are already right-continuous, adding left-continuity simply makes it continuous. Hence $\mathbb{P}(X = x) = 0$ for all x if and only if F_X is continuous. $\square$

Definition 4.3.5. A function $f\colon \mathbb{R} \to [0,\infty)$ is said to be a *probability density function* or *pdf* of a continuous random variable X if

$$\int_{-\infty}^{x} f(t)\,dt = F_X(x).$$

If it exists, the pdf of a random variable is unique.

Proposition 4.3.5 (Properties of pdfs)**.** *Let X be a continuous random variable with probability density function f_X. Then:*

1. *For all $x \in \mathbb{R}$, we have $f_X(x) \ge 0$.*
2. $\displaystyle\int_{-\infty}^{\infty} f_X(t)\,dt = 1.$
3. *For real numbers $a < b$,*
$$\mathbb{P}(a < X \le b) = \int_a^b f_X(t)\,dt.$$
4. *For almost all values of x,*
$$\frac{dF_X(x)}{dx} = f_X(x).$$

Remark 4.3.3. Importantly, **not all** continuous random variables have a pdf. These are, however, rare and pathological.

4.4. Expectation

Definition 4.4.1. Let X be a discrete random variable with pmf p_X. Then, the *expected value*, or *mean* of X is defined to be

$$\mathbb{E}[X] = \sum_{x\in S_X} x p_X(x),$$

provided that the sum converges absolutely. If the state space S_X is finite, then the expected value is always defined. The expected value is sometimes denoted μ_X or simply μ if X is clear from context.

Definition 4.4.2. Let X be a continuous random variable with pdf f_X. Then, the expected value of X is defined to be

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx,$$

provided that the integral converges absolutely.

Proposition 4.4.1. *Let X be a random variable and let $g\colon \mathbb{R} \to \mathbb{R}$ be a function.*

1. *If X is a discrete random variable with pmf p_X, then*

$$\mathbb{E}[g(X)] = \sum_{x \in S_X} g(x) p_X(x),$$

provided the sum converges absolutely.

2. *If X is a continuous random variable with pdf f_X, then*

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx,$$

provided that the integral converges absolutely.

Proof. We prove the discrete case, and leave the continuous case as an exercise. Clearly, $Y \coloneqq g(X)$ is a discrete random variable, so working from the definition, we have

$$\begin{aligned}\mathbb{E}[g(X)] = \mathbb{E}[Y] &= \sum_{y \in S_Y} y\, \mathbb{P}(Y = y) \\ &= \sum_{y \in S_Y} y\, \mathbb{P}(g(X) = y).\end{aligned}$$

Let $g^{-1}(y) \coloneqq \{x \in \mathbb{R} : g(x) = y\}$ be the preimage of y. Then, we see that $g(X) = y$ is equivalent to $X \in g^{-1}(y)$, so

$$\begin{aligned}\mathbb{E}[Y] &= \sum_{y \in S_Y} y\, \mathbb{P}(X \in g^{-1}(y)) \\ &= \sum_{y \in S_Y} y \sum_{x \in g^{-1}(y)} \mathbb{P}(X = x) \\ &= \sum_{y \in S_Y} \sum_{x \in g^{-1}(y)} y\, \mathbb{P}(X = x),\end{aligned}$$

where the swapping of the summations is justified by absolute convergence. Now, inside the sum, $y = g(x)$, and moreover, $\mathbb{P}(X = x) = p_X(x)$ by definition. Note that summing over all $y \in S_Y$ and $x \in g^{-1}(y)$ is equivalent to summing over all $x \in S_X$, since S_X is partitioned by the set of all preimages of $y \in S_Y$. Thus, we obtain

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] = \sum_{x \in S_X} g(x) p_X(x)$$

as desired. □

Proposition 4.4.2. *Let X be a random variable, and $a, b \in \mathbb{R}$ be constants. Then,*

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

Definition 4.4.3. The *variance* of a random variable X is defined by

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

This is sometimes denoted as σ_X^2 or just σ^2 if X is clear from context.

Definition 4.4.4. The *standard deviation*, denoted σ_X or σ or $\operatorname{sd}(X)$, of a random variable X is the square root of its variance. That is,

$$\sigma_X = \sqrt{\operatorname{Var}(X)} = \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}.$$

Remark 4.4.1. The standard deviation is a measure of the spread or variability of a random variable. This is because the variation measures the average squared distance to the mean. The reason why the standard deviation is the square root of the mean is to make the units equal to the units of the original random variable. The reason why the variation is the expected value of the squared distance to the mean, instead of absolute value or some other metric, is because it simplifies calculations.

Proposition 4.4.3 (Properties of variance and standard deviation)**.** *Let X be a random variable.*

1. $\operatorname{Var}(X) \geq 0$.
2. *If $Y = aX + b$ for real numbers a and b, then* $\operatorname{Var}(Y) = a^2 \operatorname{Var}(X)$ *and* $\operatorname{sd}(Y) = |a| \operatorname{sd}(X)$.
3. $\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Definition 4.4.5. The k^{th} *moment* (about the origin) of a random variable X is given by

$$\mu_k = \mathbb{E}[X^k].$$

The k^{th} *central moment* of a random variable X is given by

$$\nu_k = \mathbb{E}[(X - \mathbb{E}[X])^k].$$

Thus $\mathrm{Var}(X) = \nu_2 = \mu_2 - \mu_1^2$.

Remark 4.4.2. The first few moments give valuable information about the distribution.

- The first moment μ_1 tells you the mean.
- The second central moment ν_2 tells you the variance.
- The third central moment ν_3 (after normalisation) tells you the *skewness*, that is, how lopsided the distribution is. A random variable with a perfectly symmetric pdf will have a skewness of zero.
- The fourth central moment ν_4 (after normalisation) tells you about the heaviness of the tail.

Proposition 4.4.4 (Moment via tail probability formula)**.** *Let X be a non-negative random variable and n be a positive integer. Then,*

$$\mathbb{E}[X^n] = n \int_0^\infty x^{n-1}(1 - F_X(x))\, dx.$$

Proof. First, note that

$$x^n = \int_0^x n y^{n-1}\, dy.$$

Thus, by Proposition 4.4.1, we have

$$\begin{aligned}
\mathbb{E}[X^n] &= \int_0^\infty x^n f_X(x)\, dx \\
&= \int_0^\infty \left(\int_0^x n y^{n-1}\, dy \right) f_X(x)\, dx.
\end{aligned}$$

We may apply Fubini's theorem (or rather, Tonelli's theorem due to the non-negativity of the integrand) to exchange the order of integration. The innermost integral will be integrated with respect to x from y to ∞, because we have that

$y \le x$, and x is unbounded. The outermost integral will be integrated with respect to y from 0 to ∞. Thus, we have

$$\begin{aligned}
\mathbb{E}[X^n] &= \int_0^\infty ny^{n-1}\left(\int_y^\infty f_X(x)\,dx\right)\,dy \\
&= n\int_0^\infty y^{n-1}\,\mathbb{P}(X > y)\,dy \\
&= n\int_0^\infty y^{n-1}(1 - F_X(y))\,dy,
\end{aligned}$$

from which the formula is obtained after a changing the dummy variable y back to x. □

4.5. Discrete probability distributions

4.5.1. Bernoulli

Definition 4.5.1. A random variable X has a *Bernoulli distribution* with parameter $p \in [0,1]$ if $S_X = \{0,1\}$ and its pmf is

$$\begin{aligned}
p_X(1) &= p \\
p_X(0) &= 1 - p.
\end{aligned}$$

We say that X is a *Bernoulli random variable.*

A *Bernoulli trial* is an experiment whose outcome is modeled by a Bernoulli random variable.

Proposition 4.5.1. *If X has a Bernoulli distribution with parameter p, then*

$$\begin{aligned}
\mathbb{E}[X] &= p \\
\mathrm{Var}(X) &= p(1-p).
\end{aligned}$$

4.5.2. Binomial

Remark 4.5.1. A binomial random variable is equivalent to the number of successes in n independent Bernoulli trials.

4.5.3. Geometric

Remark 4.5.2. A geometric random variable can be thought of as the number of failures before the first success.

4.5.4. Negative binomial

Remark 4.5.3. A negative binomial random variable can be thought of as the number of failures before the rth success.

4.5.5. Hypergeometric

Remark 4.5.4. A hypergeometric random variable can thought of as the number of failures before the rth success, without replacement. For example, a hypergeometric random variable describes the number of blue balls taken from a basket of blue and red balls before taking the rth red ball.

4.5.6. Poisson

4.5.7. Uniform

4.6. Continuous probability distributions

4.6.1. Uniform

4.6.2. Exponential

4.6.3. Gamma

4.6.4. Beta

4.6.5. Pareto

4.6.6. Weibull

4.6.7. Normal

4.6.8. Cauchy

4.6.9. Lognormal

4.7. Bivariate random variables